

Exact Audits of Worked Systems for the Obstruction Calculus: Counts, Witnesses, and Monitoring Design Rules

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Abstract

Viability certificates are useful in application to the extent that their behaviour on concrete systems is known exactly. This paper develops one exact audit frame — integer and rational arithmetic throughout, every count regenerated by script — and applies it to a family of worked systems that instantiate the mechanisms of the obstruction calculus (Abaee, 2026b): a shared finite two-coordinate system with observation structures (aggregation, policy-class restrictions, decentralized observation, regime uncertainty), a review-timing grid for a hidden-regime resource model, and a continuous two-floor production benchmark. The audits yield the following. The review-timing identity on the 48-cell grid is non-strict: viability at the deadline holds if and only if $T_{\text{obs}} \leq \sigma^*(z_0) = z_0 - 1$, boundary included; the strict form fails at exactly one cell. On the four-state monitoring target, 7 of the 15 partitions are adequate with two minimal two-cell partitions, and the two maximal common-action sets overlap, so no coarsest adequate design exists. Four policy classes have kernel sizes 24, 26, 25, and 28 of 36 belief pairs — the classes a two-by-two product of aggregation level and protocol — with incomparable middle cells. Decentralized unanimity over a 256-law space sustains 12 of 36 pairs; an authority restriction collapses the kernel to zero. Mode-blind operation loses exactly the pairs containing one state. The biased certainty-equivalence law has drift at least 21/100 on $[1, 2]$; the bias-free law has drift identically zero. The benchmark’s critical aggregate is $Y^* = 27/5$, with a rational dual certificate of margin 3/50. Static duality holds at value $-1/10$ on the two-floor instance with an ℓ_1 -normalized Farkas pair, and a Helly-tight triple exhibits the sparse $m+1$ -point witness. The complete result tables are carried here: the master table of all 36 belief pairs against the five kernels, the sixteen sustaining law pairs of the decentralized protocol (an exact 4×4 product), the 48-cell timing grid, the fifteen-partition census with each fibre’s common safe-action set, and the benchmark and drift tables, with a five-figure set regenerated from the same scripts. Design rules follow: review deadlines are boundary-inclusive; merged fibres call for splits, not threshold tuning; aggregated indices support certainly-safe reporting only; biased observation maps call for structural correction.

Keywords: viability kernels; obstruction cer-

tificates; exact arithmetic; monitoring design; decentralized observation; certainty equivalence

1. Introduction

The obstruction calculus states sound sufficient conditions for nonviability — obstruction certificates — mechanism by mechanism: common-action failure, review-timing failure, tube failure, finite-time exit, certification failure through fibre crossing, and failure induced by restricting the policy class (Abaee, 2026b). Each mechanism is accompanied in application by quantitative questions. How many belief states does an observation structure destroy, and which ones? At which review deadlines does the timing certificate fire, and does it fire at the boundary? What exactly does a vote protocol cost, and which law in the protocol’s space sustains a given target? This paper answers such questions exactly for a family of worked systems, under one audit frame and one notation.

Three properties define the frame. First, all computations are carried out in integer and rational arithmetic: no solver, no tolerance, no simulation. Second, every system in the family is finite and every count is regenerated by a deterministic standard-library script, so any two executions agree. Third, each audited quantity is stated together with a witness or a certificate — an action sequence, a partition, a law pair, a dual measure — so the counts carry their own evidence. This edition carries the audit’s complete result tables and its figure set: the per-pair membership table behind the kernel counts, the sustaining law pairs behind the decentralized count, the full timing grid, the full partition census, and the benchmark and drift tables, every entry of which is regenerated from the same scripts that verify the text.

The systems are drawn from the applied model of Abaee (2026a), in which a two-coordinate resource stock is managed by acting one of two instruments under incomplete observation. The family comprises: the shared audit system with its observation structures (Sections 3.–5. and 7.–8.); the hidden-regime review grid (Section 6.); a continuous two-floor production benchmark (Section 9.); and a multiple-floor line extending the Farkas certificate to jointly binding constraints (Section 12.).

2. Systems, conventions, and methods

The shared audit system. States are pairs $x = (z_1, z_2) \in \{0, 1, 2, 3\}^2$; the safe set is $\mathcal{V} = \{z_1 \geq 1, z_2 \geq 1\}$ with the two coordinates as floors. Instruments are $u \in \{0, 1, 2\}$: $u = j \in \{1, 2\}$ acts coordinate j , which regenerates one unit up to the cap 3, while the other coordinate loses one unit; $u = 0$ executes no instrument and both coordinates lose one. The corner $(1, 1)$ is the single state from which no instrument maintains both floors, so its singleton is nonviable; the nine safe states generate $C(9, 2) = 36$ two-element belief pairs, the audited population (28 of the pairs full-information-viable; column F of Table

eftab:master), at review horizon $H = 12$ under the audited two-horizon verdict (the winning recursion at H and at $H - 2$; Abaee, 2026b, Theorem 1).

Observation structures. An observation map $\text{info} : x \mapsto c$ partitions a belief into fibres; a policy acts per fibre on the read cell. The audited policy classes combine two aggregation levels (full information; aggregation $c(x) = z_1 + z_2$) with two protocols (free per-step choice; the no-repeat protocol that forbids repeating the previous action — one previous-action register per information state, shared by all fibres, not one per branch).

Exactness. Counts are computed by backward recursion over fibre partitions with exact state arithmetic; witnesses are extracted as step-one per-fibre action vectors along a winning branch. The census of Section 7. enumerates all 15 partitions of a four-state target. The continuous objects of Section 9. are rationals.

Conventions. On the hold class of the hidden-regime model, the discrete survival supremum is $\sigma^*(z_0) = \lfloor z_0 - 1 \rfloor$, with level sets $\{z_0 \geq 1 + k\}$. Two certainty-equivalence laws are audited: an uncorrected law whose index leads the state by one step, $s \mapsto s + 0.1$, and a corrected law with the lead removed; both are read through the index $g(s) = s^2$.

Figures and tables. Every figure and every tabulated entry in this paper is produced by a generation script that first recomputes the quantity from the primitives above, in exact rational arithmetic, and asserts it before drawing or printing; the master verification script of Section 14. chains both scripts and re-derives the tabulated entries independently.

3. The master table

Table 1 is the audit’s master table: all 36 belief pairs against the five kernels — the four policy classes of Section 4. (institutional, aggregate, full-information codex, full-information) and the decentralized kernel of Section 5.. Rows are grouped by the pair’s lexicographically smaller state, written $z_1 z_2$.

The column counts are the audited kernel sizes $|\mathcal{W}_{\text{inst}}| = 24$, $|\mathcal{W}_{\text{agg}}| = 26$, $|\mathcal{W}_{\text{full-codex}}| = 25$, $|\mathcal{W}_{\text{full}}| = 28$, and $|\mathcal{W}_{\text{dec}}| = 12$. Figure 1 displays the four policy-class kernels, whose middle cells are incomparable. Two set-level facts are recorded exactly. The meet of the two middle kernels equals the institutional kernel setwise. At class level the join is the full-information class; at kernel level the union of the two middle kernels falls one pair short of the full kernel: exactly the pair $\{(1, 2), (2, 1)\}$ lies in neither middle kernel. The pair is the alternation-forced pair: under full information each branch survives by its

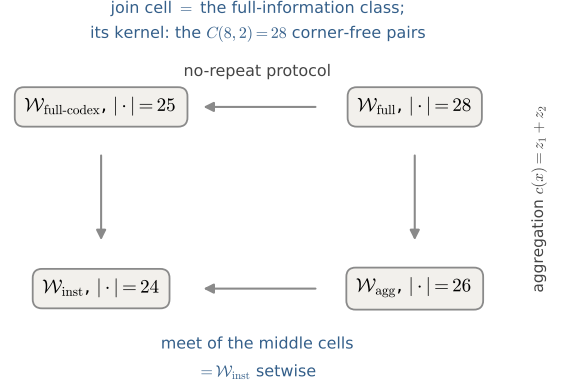


Figure 1: The four policy classes form a two-by-two product of aggregation level and protocol; the audited kernel sizes are 24, 26, 25, 28 of 36. The meet of the middle kernels is the institutional kernel setwise; the class join is the full-information class, whose kernel is the 28 corner-free pairs, one pair beyond the middle-kernel union.

own strict alternation — $(1, 2)$ under $1, 2, 1, \dots$ and $(2, 1)$ under $2, 1, 2, \dots$ — which the idealized per-state policy can carry on both branches; aggregation cannot see either requirement (one fibre, no common action), and the no-repeat protocol, whose single previous-action register must serve both branches, forbids the second branch’s required next action once the first branch’s choice occupies the register. The full kernel is exactly the $C(8, 2) = 28$ pairs avoiding the corner $(1, 1)$: the corner’s singleton nonviability accounts for every remaining pair-level loss at full information, as the eight $11|$ rows of Table 1 show.

4. Policy classes

The four audited classes combine full information and aggregation with free and no-repeat protocols. Their kernels among the 36 pairs have sizes:

$$|\mathcal{W}_{\text{inst}}| = 24, \quad |\mathcal{W}_{\text{agg}}| = 26, \\ |\mathcal{W}_{\text{full-codex}}| = 25, \quad |\mathcal{W}_{\text{full}}| = 28.$$

The four classes form a two-by-two product of aggregation level and protocol; their kernels have incomparable middle cells: the aggregate kernel contains pairs outside the full-information codex kernel (for instance $\{(1, 2), (3, 1)\}$ and $\{(1, 3), (2, 1)\}$), and the codex kernel contains pairs outside the aggregate (for instance $\{(1, 3), (3, 1)\}$). Neither protocol dominance holds. The precise meet-and-join reading is the one recorded in Section 3.. Two per-pair readings illustrate the columns of Table 1. The pair $\{(1, 2), (3, 1)\}$ is aggregate-viable but not codex-viable: the two branches read distinct fibres under aggregation, so the no-repeat protocol never binds, while at full information the winning branch sequence forces a repeated instrument. The pair $\{(1, 3), (3, 1)\}$ is codex-viable but not aggregate-viable: the two branches share the fibre $c = 4$, and no single per-fibre instrument serves both, while full information separates them.

5. Decentralized observation

Two agencies each observe one coordinate and vote; an instrument executes only on agreement. Each agency’s vote

Table 1: The master table: membership of all 36 belief pairs in the five audited kernels. I = institutional (aggregation, no-repeat); A = aggregate; C = full-information codex; F = full information; D = decentralized unanimity. A dot marks a non-member; every entry is regenerated by script.

Pair	I	A	C	F	D	Pair	I	A	C	F	D
12 21	.	.	.	F	D	22 31	I	A	C	F	D
12 31	.	A	.	F	.	22 23	I	A	C	F	.
12 23	I	A	C	F	D	22 33	I	A	C	F	D
12 33	I	A	C	F	.	22 32	I	A	C	F	.
12 22	I	A	C	F	.	23 31	I	A	C	F	.
12 32	I	A	C	F	D	23 33	I	A	C	F	.
12 13	I	A	C	F	.	23 32	I	A	C	F	D
13 21	.	A	.	F	.	31 33	I	A	C	F	D
13 31	.	.	C	F	D	31 32	I	A	C	F	.
13 23	I	A	C	F	.	32 33	I	A	C	F	.
13 33	I	A	C	F	D	11 12	.	.	.	.	.
13 22	I	A	C	F	D	11 21	.	.	.	.	.
13 32	I	A	C	F	.	11 31	.	.	.	.	.
21 31	I	A	C	F	.	11 23	.	.	.	.	.
21 23	I	A	C	F	D	11 33	.	.	.	.	.
21 33	I	A	C	F	.	11 22	.	.	.	.	.
21 22	I	A	C	F	.	11 32	.	.	.	.	.
21 32	I	A	C	F	D	11 13	.	.	.	.	.

Table 2: The 16 law pairs sustaining the audited belief $\{(1, 2), (2, 1)\}$ factor exactly: agency 1’s vote is forced to 1 at $z_1 = 1$ and 2 at $z_1 = 2$, free at $z_1 \in \{0, 3\}$; agency 2’s vote is forced to 2 at $z_2 = 1$ and 1 at $z_2 = 2$, free at $z_2 \in \{0, 3\}$. Every combination sustains — a complete 4×4 product (votes listed as $(l(z=0), l(z=1), l(z=2), l(z=3))$); agency 1’s votes are the row labels, agency 2’s the column labels.

	agency 2			
agency 1	1211	1212	2211	2212
1121	✓	✓	✓	✓
1122	✓	✓	✓	✓
2121	✓	✓	✓	✓
2122	✓	✓	✓	✓

is a map from its coordinate’s four values to the two instruments, so the law space has $2^4 \times 2^4 = 256$ law pairs. Unanimity sustains 12 of the 36 pairs (column D of Table 1), and exactly 16 coordinated-viable pairs are lost. The loss instance $\{(1, 2), (3, 1)\}$ is representative: each branch is individually sustainable, and no law pair sustains the belief. For the audited belief $\{(1, 2), (2, 1)\}$, exactly 16 of the 256 law pairs sustain it; the count coincides numerically with the 16 lost pairs, and no identity is implied.

The sustaining set has an exact product structure (Table 2): on the trajectory alternating between the branches, agency 1 is read only at $z_1 \in \{1, 2\}$ and must vote 1 then 2; agency 2 is read only at $z_2 \in \{2, 1\}$ and must vote 1 then 2; the votes at the unreachable coordinates $z \in \{0, 3\}$ are free, giving $2^2 \times 2^2 = 16$. Forbidding agency 2’s vote (fixing it to instrument 1) collapses the kernel from 12 to 0: the second agency’s voice, not the first’s, carries the audited target. Two modelling choices are declared. The deadlock transition $u = 0$ executes no instrument and both coordinates lose one; and each agency votes on its own coordinate’s current value: the audited protocol is memoryless and decentralized — no agency observes the belief, the other agency’s observation, or anything like common knowledge.

Table 3: The 48-cell timing grid: viability at the deadline against $\sigma^*(z_0) = \lfloor z_0 - 1 \rfloor$. The checkmark of column T_{obs} means the cell is viable; the identity column “id.” is the predicate $T_{\text{obs}} \leq \sigma^*(z_0)$ and column “cell” the audited viability; the two agree on all 48 cells (zero mismatches; six viable cells). The circled cell of Figure 2 is (2.0, 1): viable, on the boundary, the only cell where the strict form disagrees.

		$T_{\text{obs}} = 1$		$T_{\text{obs}} = 2$		$T_{\text{obs}} = 3$	
z_0	σ^*	id.	cell	id.	cell	id.	cell
1.0	0	.	.	.	.	.	.
1.1	0	.	.	.	.	.	.
1.2	0	.	.	.	.	.	.
1.3	0	.	.	.	.	.	.
1.4	0	.	.	.	.	.	.
1.5	0	.	.	.	.	.	.
1.6	0	.	.	.	.	.	.
1.7	0	.	.	.	.	.	.
1.8	0	.	.	.	.	.	.
1.9	0	.	.	.	.	.	.
2.0	1	✓	✓	.	.	.	.
2.1	1	✓	✓	.	.	.	.
2.2	1	✓	✓	.	.	.	.
2.3	1	✓	✓	.	.	.	.
2.4	1	✓	✓	.	.	.	.
2.5	1	✓	✓	.	.	.	.

6. Review timing on the hidden-regime grid

The hidden-regime resource model is audited on the grid $z_0 \in \{1.0, 1.1, \dots, 2.5\}$ and $T_{\text{obs}} \in \{1, 2, 3\}$, 48 cells. Viability at the deadline holds if and only if $T_{\text{obs}} \leq \sigma^*(z_0) = z_0 - 1$, with zero mismatches across the grid. The identity is boundary-inclusive: at the cell $(z_0, T_{\text{obs}}) = (2, 1)$, on the boundary $z_0 = 1 + T_{\text{obs}}$, the system is viable, and the strict form $T_{\text{obs}} < \sigma^*(z_0)$ fails at exactly this cell. Table 3 carries the full grid and Figure 2 displays it. The practical form of the rule: a review must land no later than the worst-case exit time, and a deadline exactly at the worst-case exit is admissible.

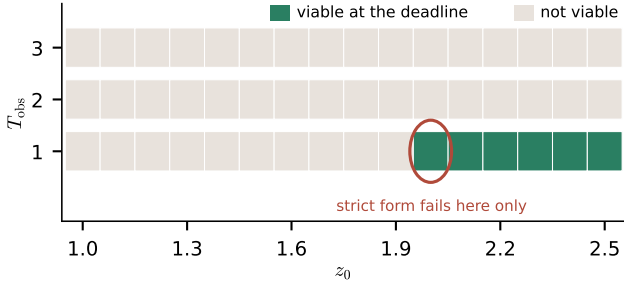


Figure 2: The timing grid. Cells viable at the deadline (green) satisfy $T_{\text{obs}} \leq \sigma^*(z_0)$; the identity is boundary-inclusive, and the strict form fails at exactly one cell, $(z_0, T_{\text{obs}}) = (2.0, 1)$ (circled).

Table 4: The fifteen-partition census on the target $\{(1, 2), (2, 1), (2, 2), (3, 1)\}$ (states written $z_1 z_2$; fibres separated by “;”). A partition is adequate when every fibre’s common safe-action set (brackets; $\cdot$ = empty) is nonempty. Seven partitions are adequate; the minimal adequate two-cell partitions are rows 6 and 7.

Partition	Adequate	Common safe actions
12 21 22 31	no	[$\cdot$]
12 21 31 ; 22	no	[$\cdot$ 12]
12 22 31 ; 21	no	[$\cdot$ 2]
12 31 ; 21 22	no	[$\cdot$ 2]
12 31 ; 22 ; 21	no	[$\cdot$ 12 2]
21 22 31 ; 12	yes	[2 1]
21 31 ; 12 22	yes	[2 1]
21 31 ; 22 ; 12	yes	[2 12 1]
22 31 ; 12 21	no	[2 $\cdot$]
22 31 ; 21 ; 12	yes	[2 2 1]
31 ; 12 21 22	no	[2 $\cdot$]
31 ; 12 22 ; 21	yes	[2 1 2]
31 ; 21 22 ; 12	yes	[2 2 1]
31 ; 22 ; 12 21	no	[2 12 $\cdot$]
31 ; 22 ; 21 ; 12	yes	[2 12 2 1]

7. Monitoring adequacy

Let $R(x)$ denote the safe actions of state x : those keeping x in $\mathcal{V}$ for one step. On the audited target, $R((1, 2)) = \{1\}$, $R((2, 1)) = \{2\}$, $R((2, 2)) = \{1, 2\}$, and $R((3, 1)) = \{2\}$. A partition of a target belief is *adequate* when every fibre has a nonempty common safe-action set. On the four-state target $\{(1, 2), (2, 1), (2, 2), (3, 1)\}$, 7 of the 15 partitions are adequate, and two minimal adequate partitions with two cells exist. Table 4 carries the complete census with each fibre’s common safe-action set; Figure 3 displays it.

The maximal subsets with a nonempty common safe-action set are $\{(1, 2), (2, 2)\}$ and $\{(2, 1), (2, 2), (3, 1)\}$. They overlap in $(2, 2)$, so no partition into maximal common-action sets exists: coarsening to maximal fibres is not an available design, and the adequate partitions found by the census are the design options. Two consequences follow. Pairwise intersection of safe-action sets is necessary for fibre adequacy but not sufficient in general; and an observation map merging states with disjoint safe-action sets obstructs at that information state while every branch remains individually viable — the remedy is a fibre split, not a finer threshold on the same index. The census makes both visible at the level of individual fibres: in each inadequate partition of Table 4 the failing fibre is exactly one merging $(1, 2)$ — whose safe-action set is the

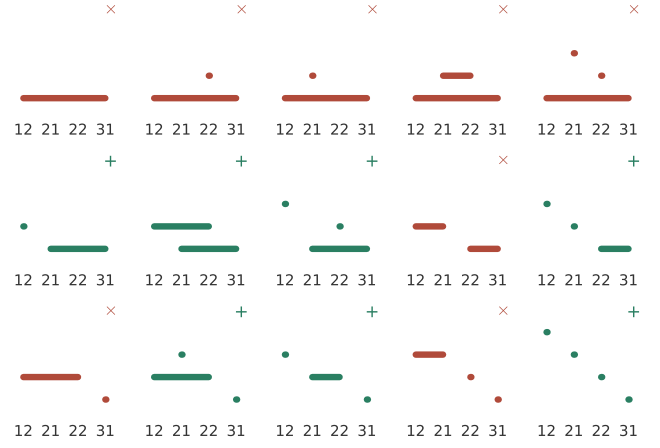


Figure 3: The census. Each panel draws one partition: a bar over a group of states is a fibre (green when its common safe-action set is nonempty, red when empty), and the margin marks adequacy (+/×). The two adequate two-cell partitions are panels 6 and 7 of the middle row.

singleton $\{1\}$ — with a state whose safe actions exclude 1, i.e. $(2, 1)$ or $(3, 1)$, and each adequate partition merges $(1, 2)$ with neither.

8. Regime uncertainty

Two regimes alternate: an adversary chooses at the end of step $t - 1$ the regime realized at step t ; the acting regime damages its coordinate by two units, the other regenerates one. With frozen regimes the kernel is $36 = 36$: the corner $(1, 1)$ is viable under each regime separately. Mode-blind operation, with the regime read only through the state, yields a kernel of exactly 28 — every pair containing the corner is lost, no merged-pair loss exists, and the loss is generated entirely at the singleton level. The mode-blind kernel agrees setwise with the full-information kernel of Section 4; the agreement is coincidental, arising under different damage routings, and no identity is claimed.

9. The continuous benchmark

Two stocks face demand floors under one control set $U = \{u \geq 0 : u_1 + u_2 \geq 2\}$ with caps $\text{cap}_1(Y) = 3/2 - (Y - 2)/10$ and $\text{cap}_2(Y) = 59/50 - (Y - 2)/10$. The critical aggregate is $Y^* = 27/5$: the cap sum equals 2 exactly there. Table 5 evaluates the caps on the audited aggregates and Figure 4 displays them. The $Y = 5$ fibre is viable with witness $(6/5, 4/5)$; the $Y = 6$ fibre is obstructed with cap sum $47/25 < 2$, certified by the dual measure $(1/2, 1/2)$ on the two active-floor atoms with margin $3/50$. The fibre width is $\Delta x_1(Y) = Y - 4$.

10. Static duality

Static duality on the two-floor instance: the max–min equals the min- λ max at value $-1/10$, with multiplier $\lambda = (1/2, 1/2)$ the ℓ_1 -normalized Farkas pair of the calculus’s worked certificate (Abaee, 2026b). Figure 5 displays the instance: under the ℓ_1 -normalized multiplier the expected drift is constant $-1/10$ on all of $[0, 1]$. Convexity is load-bearing: with $U = \{0, 1\}$ and rows $(-1, +1)$, $(+1, -1)$, the common set is empty, the max–min is -1 ,

Table 5: The benchmark’s cap arithmetic on the audited aggregates (exact rationals). The cap sum crosses the demand level 2 exactly at $Y^* = 27/5$; at $Y = 6$ the cap sum falls $3/25$ short of 2, and the ℓ_1 -normalized dual measure $(1/2, 1/2)$ certifies the obstruction with margin $3/50$ — the best feasible allocation’s expected drift equals $-3/50$.

Y	cap_1	cap_2	sum	reading
4	13/10	49/50	57/25	above
9/2	5/4	93/100	109/50	above
5	6/5	22/25	52/25	viable (witness $(6/5, 4/5)$)
27/5	29/25	21/25	2	critical
11/2	23/20	83/100	99/50	below
6	11/10	39/50	47/25	obstructed, margin $3/50$

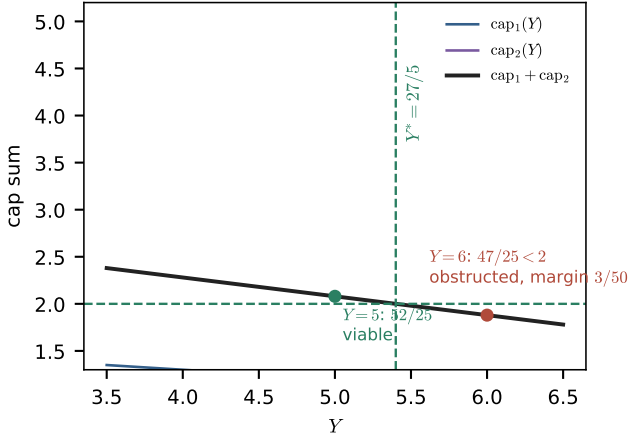


Figure 4: The benchmark. Both caps decrease linearly in the aggregate Y ; their sum crosses the demand level 2 exactly at $Y^* = 27/5$. The $Y = 5$ fibre is viable (green), the $Y = 6$ fibre obstructed with margin $3/50$ (red).

and the min-multiplier maximum is 0. Under convexity the witness is sparse: the Helly-tight triple $\{u_1 \geq 1, u_2 \geq 1, u_1 + u_2 \leq 1\}$ attains the obstruction with $m+1 = 3$ points. The feasible contrast at rows $(2/5 - u, u - 1/5)$ carries value $+1/10$ on both sides of the saddle: a certificate exists exactly where the value is nonpositive. Weak duality holds unconditionally; strong duality holds in the finite-horizon finite and static-observation classes.

11. The certainty-equivalence drift audit

For the index $g(s) = s^2$ the uncorrected law’s one-step drift is $g(s + 1/10) - g(s) = s/5 + 1/100$ on $[1, 2]$: increasing, with minimum $21/100$ at $s = 1$ and maximum $41/100$ at $s = 2$ (Table 6); the corrected law’s drift is identically zero. A biased observation map empties a nonempty kernel under an injective index, and the remedy is structural correction of the map, not a better sensor.

12. Multiple floors

With several binding floors the Farkas certificate obstructs jointly: a single certificate over all floors certifies nonviability where per-floor certificates do not compose. On the audited two-floor extension, the minimal infeasible subsystem carries explicit witnesses, and the corner $(1, 1)$ is dynamically nonviable as in the shared system. Two further objects are recorded for subsequent study: a first-breach timing bound for multi-floor systems, and a corner-cone

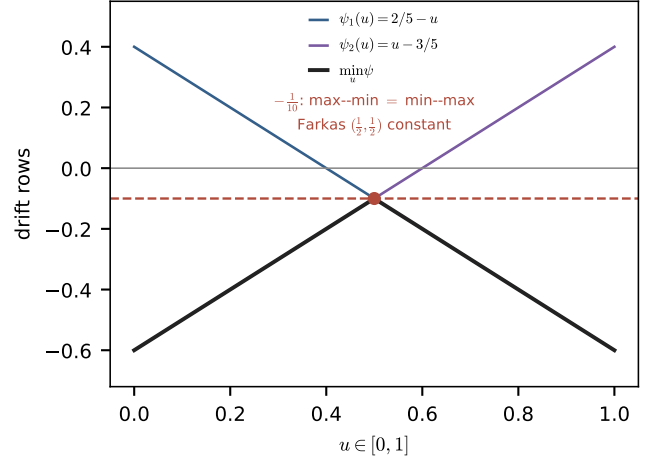


Figure 5: Static duality on the two-floor instance. The two drift rows cross at $u = 1/2$; the max–min equals the min–multiplier maximum at $-1/10$, and the ℓ_1 -normalized Farkas pair $(1/2, 1/2)$ makes the expected drift constant at the value.

Table 6: The certainty-equivalence drift audit: $g(s + 1/10) - g(s) = s/5 + 1/100$ exactly; the corrected law’s drift is 0 at every probed state.

s	uncorrected	corrected	increment per $1/4$
1	21/100	0	
5/4	26/100	0	1/20
3/2	31/100	0	1/20
7/4	36/100	0	1/20
2	41/100	0	1/20

reading of the infeasibility region.

13. Design rules

Four rules hold as identities on the audited systems and transfer as default guidance. *Timing*: reviews are boundary-inclusive — $T_{\text{obs}} \leq \sigma^*(z_0)$, the deadline at worst-case exit admissible (Section 6.). *Coarseness*: a fibre merging states with disjoint safe-action sets obstructs at that information state; the remedy is a fibre split, not a finer threshold on the same index (Section 7.). *Aggregation*: an index whose fibres cross a floor cannot support an exact safe/unsafe verdict; the certainly-safe set is the honest report. *Bias*: the uncorrected law’s drift is at least $21/100$ on a 101-point grid of $[1, 2]$, while the corrected law’s drift is identically zero; a biased observation map empties a nonempty kernel under an injective index, and the remedy is structural correction of the map, not a better sensor (Section 11.). To these the decentralized audit adds a fifth: an authority restriction on one agent’s vote can collapse the sustainable set outright, so protocol design precedes instrument tuning (Section 5.).

14. Verification methods

Every count, witness, identity, bound, figure, and tabulated entry in this paper is regenerated by deterministic scripts in exact integer and rational arithmetic, standard library only. The master verification script recomputes the counts of Sections 3.–8. from the shared system’s primitives — the transition, the fibre partition function, and the backward recursion with the no-repeat protocol

threaded as a parameter — re-derives every tabulated entry of Tables 1, 2, 3, 4, 5, and 6 independently, enumerates the census of Section 7., evaluates the benchmark identities of Section 9. in rational arithmetic, asserts the drift bounds, and checks that each headline number appears in the text. The figure-and-table generation script recomputes and asserts each plotted or printed quantity from the same primitives before drawing; the two scripts agree on every shared quantity. The audited identities are additionally covered by the per-system verification scripts of the companion papers (Abaee, 2026b; see the Code availability statement).

15. Conclusion

Exact audits bind the obstruction calculus to its worked systems: each mechanism acquires a count, a witness, and a boundary. The timing certificate fires everywhere beyond $z_0 - 1$ and nowhere at the boundary; the adequate monitoring designs on the audited target are enumerable, and no coarsest one exists; policy-class value drops unequally across aggregation and protocol restrictions, with the two middle cells incomparable, the meet of their kernels institutional setwise, and exactly one pair — the alternation-forced pair — beyond both; unanimity costs a fixed and identified set of beliefs, sustained by exactly the 4×4 product of free votes, recoverable only while both votes remain free; regime blindness loses exactly the corner’s pairs; and the continuous benchmark separates the viable from the obstructed at $Y^* = 27/5$ with rational certificates on both sides. The design rules — boundary-inclusive reviews, fibre splits over threshold tuning, certainly-safe reporting under aggregation, structural repair of bias, and protocol design before instrument tuning — are the audit’s operational content, and the master table is its evidence: every count in the paper is one column of one table, every entry of which is regenerated by script.

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Competing interests

None.

Data availability

No external data were used.

Code availability

The verification script `paper2_worked_systems_v6_verification.py` and the figure-and-table generation script `paper2_worked_systems_figures.py` are publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); they reproduce all counts, witnesses, bounds, figures, and tabulated entries verbatim.

AI declaration

GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review. The author reviewed and edited outputs and takes responsibility for the final work.